

Natural Capital and the Measurement of Productivity*

Alexander C. Abajian[†], Marc N. Conte[‡], and Eli P. Fenichel[§]

June 22, 2026

Abstract

Omitting inputs to production, such as natural capital, affects estimates of productivity growth. We show benchmark measurements of productivity growth are strongly correlated with environmental change at the national level, consistent with biased measurement due to the omission of natural capital. We derive theoretical conditions under which the measurement and inclusion of previously omitted environmental inputs reduce bias and error when estimating productivity growth. Calibrated simulations using data from the World Bank and Penn World Tables suggest including non-renewable natural capital in the measurement of TFP growth increases accuracy for 94 of the 100 countries in our sample.

Keywords: Total Factor Productivity; Natural Capital; Economic Measurement

JEL Codes: E01, O44, O47, Q56

*We thank Leonard Nakamura, Scott Wentland, Elizabeth Kiser, David Keiser, Javier Birchenall, Henry Kim, Luis Diego Herrera Garcia, Dale Manning, and seminar participants at the Yale School of the Environment for their feedback. Abajian and Fenichel gratefully acknowledge financial support from the Knobloch Family Foundation. All errors are our own.

[†]Abajian: Yale School of the Environment, alexander.abajian@yale.edu

[‡]Conte: Fordham University, marc.conte@fordham.edu

[§]Fenichel: Yale School of the Environment, eli.fenichel@yale.edu.

Innovation and technological change drove an unprecedented increase in economic activity over the past two centuries. At the same time, industrialization muddled societal awareness of fundamental connections between humans and nature. Society’s interest in tabulating the stocks of nature and their associated flows waned, and attention shifted to tracking produced and human capital. Then-President Theodore Roosevelt noted this phenomenon at the outset of the 20th century:

“With the rise of peoples from savagery to civilization, and with the consequent growth in the extent and variety of the needs of the average man, there comes a steadily increasing growth of the amount demanded by this average man from the actual resources of the country. And yet, rather curiously, at the same time that there comes that increase in what the average man demands from the resources, he is apt to grow to lose the sense of his dependence upon nature.” – [Roosevelt \(1908\)](#)

Formal systems of economic measurement followed this trend; extant structures provide little guidance for separating nature’s contribution to production from the contributions of other factors. Starting with Simon Kuznets’ report to Congress in 1934, the broader development of statistical, economic, and environmental policy has focused on measuring output and productivity growth by tracking capital and labor over time ([Kuznets 1934](#)). While governments historically emphasized tracking natural resources for posterity and avoiding overuse, features of the environment beyond land were omitted from the transition to the double entry-based national accounting systems used today. For example, sailors, vessels, and sonar are essential inputs to industrial fishing and are measured by national accountants on a regular basis; equally important is the fish stock and its ability to replenish itself over time ([Hannesson 2007](#)). However, the fish stock as an input to production is often excluded from economic measurement. This continues despite managed fish stocks (within exclusive economic zones) being a textbook example of a non-produced natural asset scheduled for measurement under modern standards like the System of National Accounts ([United Nations 2009, 2025](#)).

Here, we examine how the omission of the environment as a factor in production affects a leading measure of economic progress produced under current national accounting standards, total factor productivity (TFP). TFP growth is defined as the growth in output unexplained by growth in inputs, and it has been a benchmark metric for economy-wide technological progress and innovation for nearly 70 years ([Solow 1957](#)). The combination of TFP and a country’s capital stock provide a summary statistic for its residents’ welfare ([Basu et al. 2022](#)). Changes in TFP over time play a crucial role in shaping governments’ budget projections and policy agendas over long time horizons ([CBO 2025](#)). Economics has If natural capital affects production, then the omission of environmental assets from national accounting frameworks induces error when estimating aggregate productivity growth. Failing to account for how changes in the environment affect output may lead governments to overestimate the ability of society to grow without the use of nature or to misattribute the effects to economic policy ([Nordhaus and Tobin 1973](#); [Brandt, Schreyer, and Zipperer 2017](#)).

In this paper, we first demonstrate how omitting inputs from the production function affects residual-based estimates of TFP and discuss why natural capital is often omitted in practice. We provide predictions for the correlation of environmental change with TFP growth when natural capital is incorrectly omitted from the set of measured inputs used to construct TFP. Then, we test whether a benchmark series for TFP growth is orthogonal to proxies for natural services in a balanced panel of data for 117 countries between 1996 and 2019. We find measured growth is strongly correlated with changes in the modest set of natural capital stocks we consider. The correlations between productivity growth and natural capital persist when using alternative productivity measurements from the economics literature or those provided by national statistical agencies.

Next, we characterize when inclusion of natural capital aids measurement of TFP in theory and consider whether it does so in practice. We derive conditions under which additional measurement unambiguously reduces bias in residual-based estimates of TFP growth. These conditions prove to be restrictive and unlikely to hold in most settings. As the theoretical conditions are not formally testable, we use simulation-based evidence to gauge whether including natural capital is likely to improve measurement empirically. The results suggest that including natural capital produces gains in measurement accuracy. Across simulations calibrated to data from 100 countries, the median level of increase in accuracy, measured as the reduction in root mean squared error for estimated average TFP growth, is 7.3 basis points, or a little over 14 percent of the average TFP growth across all countries during the sample period. For the case of the United States, compounding an error reduction of this magnitude over a standard 10-year budget horizon implies a forecast error for cumulative revenues of a little over \$300 billion in 2025. This figure is two to three times larger than the annual year-on-year forecast error in deficit projections by the U.S. Congressional Budget Office (CBO), or roughly a quarter of the aggregate cost of compliance with the Clean Air Act between 1970 and 1990 (CBO 2026; U.S. Environmental Protection Agency 1997).

Literature: Tobin and Nordhaus motivated the inclusion of natural capital when accounting for growth by appealing to its amenity value (toward their broader measure of economic welfare) as well as the productive value of land (Nordhaus and Tobin 1973). Subsequent research incorporating the environment in the development accounting literature has primarily focused on improving measures of the share of national income attributable to produced capital (e.g., Mankiw, Romer, and Weil 1992; Caselli and Feyrer 2007; Monge-Naranjo, Sánchez, and Santaaulalia-Llopis 2019). These studies apportion capital income between returns on investment in produced capital and resource rents. Broadly, the literature has highlighted the importance of correcting for resource rents when comparing the returns to capital across countries while noting that the amount of residual output unexplained by factor inputs remains substantial (Hsieh and Klenow 2010). We add to this literature by characterizing how introducing previously omitted factors of production has ambiguous effects on the bias of measured output elasticities and factor shares.

A more nascent strain of literature focuses on how including natural capital affects growth accounting-based estimates of productivity. Our contribution is closest to those by Brandt, Schreyer, and Zipperer (2017) and Manning and Barbier (2025), who study the effects of adjusting existing

TFP estimates at the national level to account for the extraction of subsurface minerals and fossil fuels. Related work by Freeman, Inklaar, and Diewert (2021) highlights the importance of accounting for the fact that not all countries extract the same set of non-renewable resources domestically when comparing productivity levels across countries. This is distinct from our exercise comparing productivity growth within countries across time, but presents a methodology that would allow our calibration exercise to be applied to more granular measures of TFP. We add to this literature by explicitly characterizing the conditions under which measurement of non-renewable resources reduces errors in estimated TFP, as opposed to calculating the magnitude of the adjustments induced by inclusion. Our approach can also be generalized to consider the effects of including other potentially omitted capital stocks, such as renewable natural capital.

1 Residual-Based Productivity Measurement

Neoclassical economics attributes output to two sources: factors of production (factor inputs) and technology. Final output each instant, $Y(t)$, is determined by two objects: the level of factor inputs, denoted by the set $\mathbf{X} \triangleq \{X_1(t), \dots, X_n(t)\}$, and a summary statistic for the level of technology available to society, total factor productivity, $TFP(t)$:

$$Y(t) = TFP(t) \times F(\mathbf{X}). \quad (1)$$

TFP is a theoretical modeling device. It is an index number capturing how technology determines the production possibility frontier at a given point in time.¹ It is a parsimonious treatment for how forces such as innovation and creative destruction shift what society can produce given available resources (Romer 1990; Aghion and Howitt 1992). Under standard assumptions for the production function (e.g., the Cobb-Douglas or transcendental logarithmic forms), growth in output is given by

$$g_Y(t) = g_{TFP}(t) + \sum_{x \in \mathbf{X}} \gamma_x g_x(t), \quad (2)$$

where $g_x(t)$ terms denote growth rates and γ_x are the output elasticities for each input x at time t .^{2,3} The growth accounting equation in (2) states that all changes in output are attributable to factors of production or growth in TFP (Barro 1999). Rearranging the growth accounting equation gives

$$g_{TFP}(t) \triangleq g_Y(t) - \sum_{x \in \mathbf{X}} \gamma_x g_x(t). \quad (3)$$

1. We assume throughout this paper that technical change is Hicks neutral. A more general formulation of technical change is the partial derivative of the function determining aggregate output with respect to time.

2. Here we refer to the γ_x terms as output elasticities. Section 1 of the online appendix shows that all analysis holds when the γ_x terms are reinterpreted as the income shares for factors of production.

3. We assume throughout that output elasticities γ_x are time-invariant.

TFP growth is the output growth unexplained by growth in inputs (Comin 2010). The accounting identity in (2) requires that these two concepts of productivity growth—the growth rate of an index number for technology and the residual portion of growth unexplained by factor inputs—coincide.

In practice, TFP growth is measured using the portion of output growth not explained by growth in a set of *observed* factor inputs, which we group together as the set $\mathbf{I}$. Factor inputs (e.g., produced capital and labor) and final output are measured by national statistical agencies at regular frequencies, allowing for statisticians to take versions of (3) to the data. Residual changes in observed output—those unexplained by growth in observed inputs—are called the Solow residuals (Solow 1957) and are imputed to productivity growth by solving

$$g_A(t) \triangleq \hat{g}_Y(t) - \sum_{i \in \mathbf{I}} \hat{\gamma}_i \hat{g}_i(t), \quad (4)$$

where $\hat{g}_i$ and $\hat{\gamma}_i$ are estimates of growth and output elasticities, respectively. A Solow residual is an estimate of TFP growth. Versions of equation (4) are used by national statistical agencies to construct productivity growth in national accounts (Meyer and Harper 2005; European Commission and Eurostat 2025), and in academic applications comparing productivity across countries and over time (Feenstra, Inklaar, and Timmer 2015).

If national accountants were able to calculate the dynamics of all potential factor inputs, a natural candidate for $\mathbf{I}$ would be to include all observed stocks to ensure $\mathbf{X} \subseteq \mathbf{I}$. Therefore, a goal may be to measure all natural capital stocks and include them all at once (Nordhaus and Kokkelenberg 1999). However, in the interim, limits on institutional capacity for the scope and precision of measurement will likely cause the set of measured inputs $\mathbf{I}$ to differ from $\mathbf{X}$ and cause some error when estimating the growth rates and output elasticities of factor inputs.⁴ In the interim, the set of measured inputs $\mathbf{I}$ available to practitioners remains limited. Statistical agencies often label their versions of the estimand in (4) *multifactor* productivity (as opposed to *total factor* productivity) to acknowledge that they measure productivity growth beyond the factors they are able to account for (Organisation for Economic Co-operation and Development 2001).⁵

Environmental inputs are a leading example of factors thought to affect production that currently go excluded from residual-based approaches to productivity measurement deployed by national statistical agencies and researchers (Meyer and Harper 2005; Feenstra, Inklaar, and Timmer 2015; European Commission and Eurostat 2025; United Nations 2025).⁶ The omission of environmental inputs does not stem from active decisions to ignore nature. They instead reflect

4. It is also possible that there are natural capital stocks which produce services necessary for output which that current science is not aware of.

5. The OECD manual on productivity measurement actually states this explicitly when defining multifactor productivity: "... [t]his manual uses the MFP [multifactor productivity] acronym to signal a certain modesty with respect to the capacity of capturing all factors' contribution to output growth" (Organisation for Economic Co-operation and Development 2001).

6. At best, some TFP growth estimates adjust for payments to environmental factors that produce marketed goods and services where factor income is observed in government accounts. A unified framework would have to account for non-market services, such as the absorption of man-made pollution, when accounting for the totality of inputs into producing all goods and services humanity consumes.

limited state capacity and the difficulty of measuring how natural capital is used in production. the factor inputs in equation 1 are flows of *services* put toward production (e.g., the number of cycles by a turbine, total hours worked by the labor force, or the amount of crude oil or groundwater extracted). These flows are distinct from the stocks that create them; idle machinery, unemployed persons, and dormant subsurface minerals are not counted as inputs to production. However, in many cases measuring the totality of service flows entering production from a given resource is difficult. For example, the total harvest from a fishery in a given year may account for the harvest as an input in the fishing industry, but it will not capture the service provided by the remaining stock of fish whose reproduction helps replenish the fishery going into the next year (Hannesson 2007). Total crude oil production will capture the input of extracted petroleum, but will fail to measure how a lower level of reserves reduces underground pressurization and increases future extraction costs (Anderson, Kellogg, and Salant 2018). Measurement is further complicated by the difficulty of attributing factor income to non-market and public environmental services (e.g., precipitation and climate). In practice, changes in the flows of unmeasured services are instead imputed using changes in the capital stocks which produce them.⁷ Growth across all factor inputs can then be proxied by the change in how all stocks of resources—natural, human, or produced—are used over time.

2 Testing for Omitted Inputs

To illustrate the effects of omitting natural capital inputs when estimating TFP, consider an economy with four factor inputs: produced capital, labor, non-renewable natural capital, and renewable natural capital, such that $\mathbf{X} = \{K(t), L(t), N_1(t), N_2(t)\}$. For this economy, if a national accountant includes only produced capital and labor as inputs $\mathbf{I}$ when estimating TFP, conditional on her estimates of factor input growth rates and output elasticities $(\hat{\mathbf{g}}, \hat{\boldsymbol{\gamma}})$ the Solow residual can be decomposed as

$$g_A(t)|\mathbf{I}, \hat{\boldsymbol{\gamma}}, \hat{\mathbf{g}} = \underbrace{g_{TFP}(t)}_{\text{True innovation}} + \underbrace{\sum_{x \in \{N_1, N_2\}} \gamma_x g_x(t)}_{\text{Omitted inputs}} + \underbrace{\sum_{i \in \{K, L\}} [\gamma_i g_i(t) - \hat{\gamma}_i \hat{g}_i(t)]}_{\text{Measurement error}}. \quad (5)$$

When natural capital inputs are omitted and $\mathbf{X} \setminus \mathbf{I} \neq \emptyset$, (3) will in general fail to be an unbiased estimator for TFP growth; the contributions of the omitted natural capital inputs are misattributed to TFP because they are spuriously absorbed by the Solow residual.⁸ Correlation of the Solow residual in (6) with changes in a given natural capital stock (or other potentially-omitted input) over time is evidence that the stock in question may be an omitted input in production. Conversely,

7. This imputation is further complicated by the fact that the levels of stocks typically influence market or shadow prices of services entering production.

8. It is of course possible that the sum of the omitted input and measurement error terms are zero in expectation and the estimator is unbiased even when inputs are omitted. We view this case as unlikely.

a precise null of zero correlation between a natural capital stock and estimated TFP growth suggests that the empirical growth accounting equation contains the true set of factor inputs (i.e., $\mathbf{X} \subseteq \mathbf{I}$), and omitting the candidate stock for inclusion is unlikely to bias measures of TFP growth.

The decomposition in (6) shows three channels by which residual-based estimates of TFP growth can be correlated with natural capital: (1) correlation between natural capital use and unobservable growth in TFP, (2) correlation with omitted inputs that affect output (in this case, the natural capital stocks which are elements of $\mathbf{X}$ and omitted when estimating TFP growth), and (3) correlation with measurement error when calculating growth rates or output elasticities of included factors. While an econometrician cannot distinguish among these three potential sources of correlation, she can test whether the Solow residuals are uncorrelated with changes in natural capital stocks. If growth in natural capital is correlated with measured TFP growth, then we cannot rule out channel (2). Finding this form of empirical correlation in the data suggests that natural capital is more likely to play a role in output.

2.1 Omitted Factors in Practice

In a world with perfect information on the growth rates of factor inputs and TFP, we could falsify that a given natural capital stock is an omitted variable by testing whether a residualized version correlates with TFP (channel 2). However, as we only observe g_A jointly as the sum of the three terms in (6), we are restricted to testing the correlations between growth in *measured* natural capital inputs and TFP. As mentioned in the introduction, the twin difficulties of estimating TFP growth are that the set of true factor inputs is unknown to the econometrician and that even when factors are known in theory, and that measurement of individual services is difficult due to limits on state capacity. This renders falsifying individual services that go unmeasured infeasible.

For concreteness, consider a case of the four input example above that the renewable natural capital input, N_2 , stems solely from fisheries, which provides a service with output elasticity γ_2 each year. This service encompasses the biomass extracted by commercial marine harvest operations as well as food web regulation, nutrient cycling, and serving as a pump for underwater substrates (Holmlund and Hammer 1999). Suppose the Solow residual is constructed on an annual basis by subtracting year-over-year growth in (elasticity-weighted) capital, labor, and non-renewable resources from growth in output. The discrete analog of (6) becomes

$$g_{A,t} = g_{TFP,t} + \gamma_2 g_{N_2,t} + \sum_{i \in \{K,L,N_1\}} [\gamma_i g_{i,t} - \hat{\gamma}_i \hat{g}_{i,t}] \quad (6)$$

The difficulty facing an econometrician seeking to rule out fisheries as an omitted input is the ecosystem services beyond harvested biomass are unmeasured in practice or difficult to quantify a residual including fisheries is that it's difficult. While we cannot directly measure all ecosystem services fisheries provide each year, the stock of fish in the fishery (as opposed to all productive services produced) is often measured directly at an annual (i.e., discrete) frequency. Suppose we regress the (demeaned) Solow residual on year-over-year growth in the fish stock, denoted $g_{S_2,t}$

$$g_{A_t} = \beta_2 g_{S_{2,t}} + \varepsilon_t. \quad (7)$$

By Frisch-Waugh-Lovell, the population coefficient β_2 in the projection of annual Solow residuals on year-over-year fish stock growth in (7) identifies

$$\beta_2 = \frac{\text{Cov}\left(\left(g_{TFP,t} + \gamma_2 g_{N_{2,t}} + \sum_{i \in \{K,L,N_1\}} [\gamma_i g_{i,t} - \hat{\gamma}_i \hat{g}_{i,t}]\right), g_{S_{2,t}}\right)}{\text{Var}(g_{S_{2,t}})}. \quad (8)$$

When the measurement error terms and underlying TFP growth are (conditionally) uncorrelated with changes in the fish stock, the estimand β_2 becomes

$$\beta_2 = \gamma_2 \frac{\text{Cov}(g_{N_{2,t}}, g_{S_{2,t}})}{\text{Var}(g_{S_{2,t}})}. \quad (9)$$

Thus as long as changes in fish stocks are a valid instrument for changes in the services they provide (i.e., $\text{Cov}(g_{N_{2,t}}, g_{S_{2,t}}) \neq 0$, $\text{Cov}(g_{TFP,t}, g_{S_{2,t}}) = 0$, and $g_{S_{2,t}}$ is uncorrelated with measurement error), the output elasticity of services from fisheries is identified up to a multiplicative constant.⁹ The assumptions required to go from (8) to (9) induces threats to identification. First, there is potential for a form of type one error where we correctly reject the null hypothesis of no correlation between g_A and g_{N_i} , but spuriously attribute it to N_i being an omitted input in production instead of some combination of channels 1 (correlation with unobservable TFP growth), 2 (correlation with other omitted inputs), or 3 (correlation with errors in measurement). This also brings about the potential for a form of type two error: when g_{N_i} is an omitted variable in the growth accounting equation and correlated with other omitted inputs, measurement error, and/or unobservable TFP growth, these correlations may offset and lead the econometrician to (correctly) fail to reject the null hypothesis of no correlation between g_A and g_{N_i} , but spuriously attribute this to natural capital playing no role in production.¹⁰

2.2 Identification in the General Case

In this general case, we have access to measured growth in a set of resource stocks $\mathbf{S}$ to use as instruments for a potentially unknown number of omitted factors $\mathbf{X} \setminus \mathbf{I}$ which enter $g_{A_{j,t}}$. TFP from the PWT $g_{A_{j,t}}$ is constructed by residualized growth in capital and labor inputs weighted by their measured factor shares under the assumption of CRS production.¹¹ Through the lens of our

9. For this estimand to be unconditionally well-defined we also require $(g_{N_{2,t}}, g_{S_{2,t}})$ to be jointly covariance stationary.

10. Several features of measurement may bring about these spurious results in practice. First, estimating output elasticities is nontrivial even for cases where factor income can be reasonably attributed (Caselli and Feyrer 2007). Second, the lack of attributed factor income makes it difficult to delineate which changes in measured natural capital stocks constitute proxies for changes in flows entering production. Finally, while measures of some forms of innovation exist (e.g., patent filings and seed varietals, used as proxies for innovation in Acemoglu et al. (2023) and Moscona and Sastry (2023) respectively), underlying correlations between environmental change and TFP growth are not falsifiable.

11. See equations C4 and C5 in Appendix C of Feenstra, Inklaar, and Timmer (2015) for details.

framework, the set of included inputs is $\mathbf{I} = \{K, L\}$ and $g_{A_{j,t}}$ can be decomposed as

$$g_{A_{j,t}} = g_{TFP_{j,t}} + \sum_{x \in \mathbf{X} \setminus \mathbf{I}} \gamma_x g_{i,j,t} + \sum_{i \in \{K,L\}} [\gamma_i g_{i,j,t} - \hat{\gamma}_i \hat{g}_{i,j,t}]. \quad (10)$$

In the version we take to the data, we instead consider specifications of the form

$$g_{A_{j,t}} = \sum_{i \in \mathbf{I}} \beta_s g_{S_{i,j,t}} + \boldsymbol{\theta}' \mathbf{Z}_{j,t} + \varepsilon_{j,t} \quad (11)$$

where the outcome is $g_{A_{j,t}}$ the residual-based measurement of TFP growth in country j in year t taken from the Penn World Tables (Feenstra, Inklaar, and Timmer 2015), $S_{i,j,t}$ are measurements of changes in natural capital stocks, and $\mathbf{Z}_{j,t}$ is a vector of controls. In this general case, the included natural capital stocks $S_{i,j,t}$ are intended to jointly identify the true underlying natural capital inputs.

The population coefficients β_s from regressing $g_{A_{j,t}}$ onto the set of included stocks and any other controls identifies

$$\beta_s = \frac{\text{Cov}(g_{A_{j,t}}, \tilde{g}_{S_{i,j,t}})}{\text{Var}(\tilde{g}_{S_{i,j,t}})} \quad (12)$$

where $\tilde{g}_{S_{i,j,t}}$ are the residuals from a regression of $g_{S_{i,j,t}}$ from all other covariates and a constant. Substituting (10) into numerator of (11) gives:

$$\text{Cov}(g_{A_{j,t}}, \tilde{g}_{S_{i,j,t}}) = \text{Cov}\left(g_{TFP_{j,t}} + \sum_{x \in \mathbf{X} \setminus \mathbf{I}} \gamma_x g_{x,j,t} + \sum_{i \in \mathbf{I}} [\gamma_i g_{i,j,t} - \hat{\gamma}_i \hat{g}_{i,j,t}], \tilde{g}_{S_{i,j,t}}\right) \quad (13)$$

when the stocks instrumenting omitted services are uncorrelated with measurement error and underlying TFP growth, we have

$$\beta_s = \sum_{x \in \mathbf{X} \setminus \mathbf{I}} \gamma_x \frac{\text{Cov}(g_{x,j,t}, \tilde{g}_{S_{i,j,t}})}{\text{Var}(\tilde{g}_{S_{i,j,t}})} \quad (14)$$

Regression coefficients β_s identify a weighted sum of output elasticities associated with the omitted capital stocks where some weights may be negative.

2.3 Inference

The β_s terms are the estimands of interest. While the sign on these coefficients is ambiguous and depends on the weighted summations over correlations with unobservable service inputs in (14), our prior is that the coefficients for these six natural capital stocks will be positive. For instance, extraction from fisheries and timber resources or larger amounts of power produced by hydroelectric generation should result in greater output. A larger area of land available for economic activity

allows for more structures, agriculture, recreation, and inhabitants. Thus, we expect an expansion in any of these natural capital stocks to be positively correlated in growth with output and in turn TFP when they are omitted from its construction. However, the correlation between natural capital inputs and TFP need not be positive generally; they will instead depend on physical measurement. Changes in environmental services such as heat during the summer and floods may instead be negatively correlated with output as they capture degradation in the underlying natural capital in the form of the climate (Lemoine, Hausman, and Shrader 2026).

Inference on these coefficients does not test for causal effects from changes in natural capital stocks on productivity growth, nor do these coefficients separately identify the omitted variable bias we seek to test from other potential channels for correlation. Rather, these coefficients identify the (conditional) covariance between growth in natural capital stocks and growth in measured TFP. Failing to reject the joint null hypothesis of interest:

$$H_0 : \beta = \mathbf{0} \tag{15}$$

only indicates that we cannot rule out that the set of natural capital stocks we consider is uncorrelated with the combination of components in the decomposition of TFP growth in (6).

Three assumptions must hold in order to interpret the rejection of the null hypothesis in (15) as evidence that natural capital inputs are omitted when calculating TFP. First, measurement error in the (weighted) factor input growth terms used to construct estimates of TFP is uncorrelated with growth in natural capital stocks. Second, changes in natural capital stocks are a good proxy for changes in the flow of services they produce; they are strongly correlated with the services they are intended to instrument and uncorrelated with any other omitted factor inputs beyond the natural capital stocks we consider. Third, the rate of true technical change g_{TFP} is conditionally uncorrelated with changes in environmental inputs. These assumptions are strong. However, their failure would not indicate that natural capital is irrelevant for national accountants. Rather, if the first assumption fails, then better measurement of natural capital improves the measurement of factor inputs that currently enter national accounts even if measurement of natural capital provides no direct value when calculating productivity growth. If the second assumption fails, then it is a powerful argument for adjusting the natural capital stocks we measure in order to keep better track of the aspects of the environment that are important for the economy. Finally, if the third assumption fails, then measurement of natural capital is a first-order priority for national accountants in order to determine how nations may adjust the use of the environment to accelerate true productivity growth.

2.4 Data on Natural Capital

Table 1, describe covariates and controls, and per Scott Wentland add a hidden service we believe they identify

To test for correlations between natural capital and measured productivity, we examine the

relationship between measured national TFP growth from the Penn World Tables (PWT) version 10.01 (Feenstra, Inklaar, and Timmer 2015) and growth in the six country-level natural capital stocks measured by the 2024 version of the World Bank’s The Changing Wealth of Nations (CWON) dataset.

Columns one and two of table 1 show the six renewable capital stocks in the CWON that will our set of environmental inputs we use as covariates along with the native units of measurement for quantities over time. The timber land variable is constructed based on the total hectares of productive timber area at the country level as measured by the Food and Agriculture Organization of the United Nations (FAO). The agricultural land variable is constructed using data from the World Bank’s World Development Indicators series based on total crop and pasture area measured by the FAO (World Bank 2024a). Non-timber forest cover measures the total area of forests providing ecosystem services beyond wood products and is intended to capture the change in “... the broader ecological contributions of forests, including carbon storage, water regulation, and the provision of non-wood forest products (NWFP)” over time (World Bank 2024a). CWON data on mangrove ecosystem coverage are taken from the Global Mangrove Watch (1996 - 2020) Version 3.0 Dataset (Bunting et al. 2022). Data on annual fisheries biomass were provided to the authors of the CWON by researchers at the Sea Around Us research initiative at the University of British Columbia (Pauly, Zeller, and Palomares 2020). Finally, the CWON data on annual hydroelectric power generation—electricity generated, not capacity—are taken from Midsummer Analytics (World Bank 2025).

Column three of Table 1 gives the unmeasured environmental services we believe are identified by changes in each renewable capital stock. For example, agricultural land provides nutrients for crops and grazing space for animal husbandry, non-timber forests provide recreation, mangroves protect produced capital from floods, and hydroelectric power provides electricity. Increases in the quantities of these natural capital *stocks* in column one are likely associated corresponding increases in these services. This connection also highlights the need to use the practical need for using stocks as proxies for services; while resource extraction such as timber harvest and total landed catch are often tracked by statistical agencies, they are omitted as inputs from existing calculations of TFP (Feenstra, Inklaar, and Timmer 2015). Many other ecosystem services that are recognized to affect production (e.g., water purification from forests, flood prevention from mangroves, and nutrient cycling by fisheries) go unmeasured entirely in many countries, rendering their inclusion in TFP calculations infeasible.

Column four of Table 1 gives the potential confounders associated with the natural capital stocks used as instruments. While in a perfect world the stocks would only correlate with the omitted services we hope to identify in Column 3, they may also be correlated in the opposite direction with additional omitted inputs. As shown in equation (14), β_s is in general a non-convex combination of the output elasticities associated with omitted factor inputs. When the covariances $\text{Cov}(S_s, N_n)$ do not all (weakly) share the same sign across the underlying omitted inputs N_n , the estimand in (14) can be zero even when the set of omitted variables contains inputs for which S_s is

a good instrument. For example, an increased fishery landing or timber harvest also decreases the level of ecosystem services provided by the harvested biomass.

Finally, as the CWON dataset in its raw form contains an unbalanced panel of natural capital stocks in each country covering 1995-2019, columns five and six show the coverage of each of the six natural capital stocks in terms of total missing entries as well as the number of countries (out of the 117 we consider).. In general, cover of land assets is quite good while coverage of mangroves, fisheries, and hydroelectric generation is more sparse. When taking these data to the the regression model, we impute missing values as zero changes from the last period where measurements were available, leaving us with a balanced panel of growth in natural capital stocks between 1996 and 2020. Sections 1.2 and 1.3 of the online appendix give a discussion of the imputation process for missing data along with robustness checks to ensure the results are not an artifact of this process.

2.5 Estimation

Our baseline empirical specification is a standard two-way fixed effects model where we regress measured TFP growth at the country level ($g_{A_{j,t}}$) on growth in the six natural capital stocks ($g_{S_{n,j,t}}$) in Table 1. Equation (16) shows the empirical analog of (6) that we take to the available data. We estimate (16) on a balanced panel of 117 countries over the 1996-2019 period:

$$g_{A_{j,t}} = \sum_{s \in \mathbf{S}} \beta_s g_{S_{s,j,t}} + \theta' \mathbf{Z}_{j,t} + \delta_j + \zeta_t + \varepsilon_{j,t} \quad (16)$$

where j and t index countries and years respectively and the set $\mathbf{S}$ covers the six natural capital stocks. Coefficients β_s govern their relationship with TFP growth. $\mathbf{Z}_{j,t}$ is a vector of country-level characteristics each year that controls for inputs such as produced capital, employment, labor's share of income, human capital taken directly from the PWT as well as idiosyncratic time trends. Terms (δ_j, ζ_t) allow for country and year fixed effects (labeled TWFE in Table 1), and $\varepsilon_{j,t}$ capture residual idiosyncratic variation in TFP growth.

2.6 Results

Table 2 displays estimates of $\hat{\beta}$ from various specifications of (16), how expected growth in TFP varies with changes in the growth of a given natural capital stock. Each column gives two tests of the null hypothesis in (15) that the vector of coefficients on growth in natural capital $\boldsymbol{\beta} = [\beta_1, \dots, \beta_7]'$ is the zero vector. The bottom rows of table 2 show p -values from two Wald tests of this joint null hypothesis, one using asymptotic standard errors and a second using a semi-parametric wild bootstrap approach (Wu 1986). We can reject that the natural capital stocks we consider are uncorrelated with productivity growth at traditional significance levels under the tests using asymptotic standard errors across all specifications, and the tests using the wild bootstrap approach are always statistically significant at the 10 percent level. Almost all point estimates are positive, the expected sign if the natural capital stocks we consider are beneficial inputs for

production.¹²

Longer discussion of robustness checks here.

Robustness: The results in Table 2 also hold under more systematic approaches to model selection. p -values for asymptotic and wild-bootstrap tests of the joint null hypothesis in (15) hover around the 5 percent significance level for the when (16) is estimated using LASSO-based model selection methods proposed in Belloni, Chernozhukov, and Hansen (2014). The benchmark post-double-selection approach produces p -values of 0.07 and 0.01 for the asymptotic and semi-parametric tests respectively. A randomization-inference based test of joint null hypothesis in Section 1.3.1 of the online appendix, which tests the frequency at which the F -statistic under asymptotic errors is larger than the value produced by the observed data when the data are resampled under a true null hypothesis produces a p -value of 0.018.

Individual tests of coefficient precision suggest the services from land (especially those for agricultural) are the inputs most likely to affect measured TFP. To ensure missing land rents are not the sole drivers of the results here, we re-estimate equation (16) after excluding agricultural, and timber land from the set of natural capital stocks we include as covariates. Under the nonland specification, our estimates lose precision and we are only able to reject the null hypothesis at the 10 percent level (see Table 3 in the online appendix). This confirms the suggestion in Feenstra, Inklaar, and Timmer (2015) that better accounting for land rents in national accounts is a fruitful direction for future work. Online appendix 1.3.3 shows our results also hold when using alternative measures of productivity growth produced by government statistical agencies as outcome variables. Appendix Tables 4 and 5 show our results hold when we add additional natural capital stocks in the form of non-renewable resources and urban land as covariates when estimating (16). Table 6 shows the results are also present when we instead use an alternative measure of multifactor productivity for a panel of countries in the EU plus Norway taken from Eurostat. Appendix Tables 9 and 10 show that we can also reject the joint null in equation (16) when the outcome variables is replaced with real or PPP output at the national level and controls are included for produced capital, labor, and other factor inputs.

3 The Marginal Inclusion of Potentially Omitted Inputs

Section 2 provides suggestive evidence that natural capital is an omitted input in existing measures of TFP. We now turn to examining whether adjusting these measured to account for previously omitted inputs improves measurement. From here forward, we operate under the assumption that the sequence of TFP growth terms, $\{g_{TFP}(t)\}$, is independent of growth in factor inputs and that all measurement of input growth is done without error. In this case, if a national accountant had

12. The negative coefficient on forest cover may be due to increases coinciding with declines in logging or forestry activities which are counted in final output. This does not per se imply that forests are a drag on the economy, but is rather indicative that the services they produce (outside of timber) are not captured by measured output. Generally, any societal benefit produced by a given natural capital stock that is conditionally uncorrelated with market output will not cause a correlation with TFP growth in this exercise.

perfect information, the Solow residual in (4) is unbiased and consistent estimator for $g_{TFP}(t)$ when $\mathbf{I} = \mathbf{X}$ and output elasticities are known.¹³

How does this change when measurement of factor inputs is incomplete? Consider an economy where the set of inputs in aggregate production $\mathbf{X}$ comprises labor, capital, and at least $J \geq 1$ additional factors which we will refer to as (omitted) natural capital stocks indexed by $\{N_j(t)\}_{j=1}^J$. Suppose a national accountant measures growth in output, produced capital, and labor inputs. Setting $\mathbf{I} = \{L(t), K(t)\}$, with estimated output elasticities for produced capital and labor ($\{\hat{\gamma}_K, \hat{\gamma}_L\}$) in hand she can construct a baseline version of the Solow residual:

$$\hat{g}_A(t) \triangleq g_Y(t) - \hat{\gamma}_K g_K(t) - \hat{\gamma}_L g_L(t) \quad (\text{Model 1})$$

that accounts for output growth unexplained by produced capital and labor. We'll refer to this estimator as Model 1. Let $\varepsilon \triangleq \hat{g}_A - g_{TFP}$ shorthand the difference between the Solow residual and underlying TFP growth. When measured factor growth is done without error, the bias of the estimator in Model 1 conditional on estimated output elasticities is:¹⁴

$$\mathbb{E}[\varepsilon | \{\hat{\gamma}_K, \hat{\gamma}_L\}] = [\gamma_K - \hat{\gamma}_K] \mathbb{E}[g_K] + [\gamma_L - \hat{\gamma}_L] \mathbb{E}[g_L] + \sum_{j=1}^J \gamma_j \mathbb{E}[g_{N_j}] \quad (17)$$

where the $\mathbb{E}[g_{N_j}]$ s are the average expected growth rates for the omitted natural capital inputs. Let $\text{MSE}(1)$ denote the mean squared error (MSE) for Model 1.

Now, suppose the national accountant is also able to measure the use of the first natural capital input and its growth over time g_{N_1} . She estimates a separate set of output elasticities associated with the new set of inputs to aggregate production, $\{\tilde{\gamma}_K, \tilde{\gamma}_L, \tilde{\gamma}_1\}$. Her alternative Solow residual that accounts for the first natural capital input is:

$$\tilde{g}_A(t) \triangleq g_Y(t) - \tilde{\gamma}_K g_K(t) - \tilde{\gamma}_L g_L(t) - \tilde{\gamma}_1 g_{N_1}(t) \quad (\text{Model 2})$$

We refer to this alternative estimator that is inclusive of the first natural capital stock as Model 2. Letting $\delta \triangleq \tilde{g}_A(t) - g_{TFP}(t)$, the conditional bias for the Solow residual that includes the first natural capital stock is

$$\mathbb{E}[\delta | \tilde{\gamma}_K, \tilde{\gamma}_L, \tilde{\gamma}_1] = \mathbb{E}[\gamma_K - \tilde{\gamma}_K] \mathbb{E}[g_K] + \mathbb{E}[\gamma_L - \tilde{\gamma}_L] \mathbb{E}[g_L] + \mathbb{E}[\gamma_1 - \tilde{\gamma}_1] \mathbb{E}[g_{N_1}] + \sum_{j=2}^J \gamma_j \mathbb{E}[g_{N_j}] \quad (18)$$

Let $\text{MSE}(2)$ denote the mean squared error for Model 2.

The marginal inclusion of natural capital when estimating TFP using residual-based methods

13. We additionally assume throughout that the data-generating processes for growth in inputs and TFP are stationary and that the relevant moments for these processes exist such that all estimands are well-defined.

14. Going from (Model 1) to (17) below implicitly assumes that the expectations of $\tilde{g}_L$ and $\tilde{g}_K$ are mean-independent of estimated output elasticities. This assumption, unlikely to hold in practice, is unnecessary for the results below and is used only to avoid the additional notation required for conditioning expectations.

amounts to switching from Model 1 to Model 2. The biases and mean squared errors for each estimator depend on output elasticities and the expected growth rates of inputs. We now show when the bias of TFP growth estimates can be signed, and examine conditions under which including an omitted input on the margin improves measurement.

Our first result is formed when the production function displays constant returns to scale (CRS) and factor markets are competitive. Under these assumptions, the shares of national income accruing to factors are equal to their marginal products (and in turn, output elasticities) when the production function is log-linear. The income-based approach is the workhorse method for estimating factors' marginal products in aggregate production (Caselli and Feyrer 2007; Monge-Naranjo, Sánchez, and Santaaulalia-Llopis 2019). Solow residuals generated by Model 1 have the following properties.

Proposition 1. When production exhibits CRS, if $\min_j \{\mathbb{E}[g_{N_j}]\} > \max\{\mathbb{E}[g_K, g_L]\}$, then $\mathbb{E}[\varepsilon] > 0$ and Model 1 overestimates TFP growth. Conversely, if $\max_j \{\mathbb{E}[g_{N_j}]\} < \min\{\mathbb{E}[g_K, g_L]\}$, then $\mathbb{E}[\varepsilon] < 0$ and Model 1 underestimates TFP growth. In either case, going from Model 1 to Model 2 unambiguously reduces the magnitude of bias.

Proof. See Section 6.1. □

If growth in included factors is much larger or much smaller than growth in omitted factors, the bias the Solow residual can be signed. However, given the measurement of growth in omitted factor inputs likely goes unobserved, the conditions are difficult to test in practice. When the conditions in Proposition 1 fail to hold, instead we have:

Corollary 1. When growth in factor inputs does not meet either condition in Proposition 1, unless $J = 1$ the bias of Model 1 and the effects of going from Model 1 to Model 2 on this bias are ambiguous. The ambiguity remains even if output elasticities for all included factors are measured without error.

Proof. See Section 6.2. □

Unless going from Model 1 to Model 2 leads to no remaining omitted inputs in the Solow residual (i.e., when $J = 1$), it is generally ambiguous whether accounting for omitted inputs on the margin reduces bias when estimating TFP. Thus in the broadest sense, additional measurement of factor inputs and apportioning of income they generate does not guarantee improved measures of productivity growth.

The implications of the results above for the case of including non-renewable resource rents when calculating TFP growth empirically proves illustrative. Growth accounting research has long recognized that two-factor models of production are inappropriate for oil-producing economies (e.g., Mankiw, Romer, and Weil 1992). A national accountant may in principle adjust her residual-based TFP estimate by going from a baseline model inclusive only of produced capital and labor (Model 1) to one that also includes growth in the use of non-renewable resources, N_1 henceforth, that

resembles Model 2. In research settings, this has been put into practice by splitting the factor income apportioned to produced capital under Model 1 between produced and natural capital in Model 2 (e.g., Caselli and Feyrer 2007; Brandt, Schreyer, and Zipperer 2017; Monge-Naranjo, Sánchez, and Santaaulalia-Llopis 2019; Manning and Barbier 2025). Under the assumption of CRS production and competitive factor markets, an estimate inclusive of natural capital can be constructed by setting $\tilde{\gamma}_1$ to share of national income accruing to privately-owned hydrocarbon stocks and then recalculating the output elasticity of produced capital as one less the income shares for labor and natural capital, $\tilde{\gamma}_K = 1 - \tilde{\gamma}_L - \tilde{\gamma}_1$. The growth term for non-renewable resources $g_{N_1}(t)$ can be constructed as the change in the extraction of physical resources over time.

Corollary 1 implies that going from Model 1 to Model 2 is not guaranteed to improve measurement of TFP. Even in the extreme cases where either income is correctly apportioned to the new natural capital stock ($\tilde{\gamma}_1 = \gamma_1$) or the CRS assumption is relaxed and somehow the unobservable γ s for all included factors are known, unless the conditions in Proposition 1 hold, including measurement of non-renewables on the margin does not necessarily lower the bias of estimated TFP relative to Model 1. The resulting ambiguous effects on absolute bias translate to other commonly used metrics for the goodness of fit of estimates. The negative result above translates into another commonly used measure, mean squared error:

Proposition 2. When $J \geq 2$ it is ambiguous whether $\text{MSE}(2)$ is lower than $\text{MSE}(1)$ even if output elasticities for all included inputs are measured without error under both models.

Proof. See Section 6.3. □

Proposition 2 presents the non-monotonicity result which motivates this paper. Only when all growth terms (weakly) share the same sign in expectation and all output elasticities are measured without error is it guaranteed that that $\text{MSE}(2)$ is no larger than $\text{MSE}(1)$. This holds for generalized versions of Model 1 and Model 2 (e.g., when Model 2 includes more than one previously-omitted input on the margin) unless going from Model 1 to Model 2 makes it so the Solow residual is correctly specified.¹⁵

An alternative formulation for the bias in these estimators is to express the omitted variables as deviations from average growth across all natural capital factors that are omitted from a given model. Let $\mathbf{J} \subset \mathbf{X}$ be the set of omitted factors under Model 1 and $\mathbf{J}' \subset \mathbf{J}$ be the set of factors omitted under Model 2. The average factor shares and growth rates across these factors are

$$\bar{\gamma}^{\mathbf{J}} \triangleq \frac{1}{|\mathbf{J}|} \sum_{j \in \mathbf{J}} \gamma_j \quad , \quad \bar{g}^{\mathbf{J}} \triangleq \frac{1}{|\mathbf{J}|} \sum_{j \in \mathbf{J}} \mathbb{E}[g_{N_j}]$$

such that for Model 1 the bias is

$$\mathbb{E}[\varepsilon|\mathbf{J}] = |\mathbf{J}| \left(\text{Cov}(\gamma_j, \mathbb{E}[g_j] | j \in \mathbf{J}) + \bar{\gamma}^{\mathbf{J}} \bar{g}^{\mathbf{J}} \right) \quad (19)$$

15. Even this weak result may fail when some factor inputs are “bads” in the sense that higher levels reduce output (e.g., the case global mean surface temperatures).

and for Model 2 the bias is

$$\mathbb{E}[\delta|\mathbf{J}'] = |\mathbf{J}'| \left(\text{Cov}(\gamma_j, \mathbb{E}[g_j] | j \in \mathbf{J}') + \bar{\gamma}^{\mathbf{J}'} \bar{g}^{\mathbf{J}'} \right) \quad (20)$$

When the growth rate of natural capital inputs are iid (and in turn, uncorrelated with output elasticities), the covariance terms above become zero and a sufficient condition for declining absolute bias becomes

$$|\bar{\gamma}^{\mathbf{J}} \bar{g}^{\mathbf{J}}| \geq |\bar{\gamma}^{\mathbf{J}'} \bar{g}^{\mathbf{J}'}| \quad (21)$$

The converse to Proposition 2 when input growth is independent of output elasticities is that there is always a possibility for better measurement to lower bias and MSE. All combinations of additional measurement—going from omitting $\mathbf{J}$ to $\mathbf{J}'$ —where including additional inputs reduces the produce of the average growth and output elasticity of omitted factors (in absolute terms) lowers bias and MSE when input growth is independent.

The result that the marginal inclusion of natural capital need not reduce bias in estimates of TFP growth is, to our knowledge, has gone unrecognized in the growth accounting literature. For example, existing work by Brandt, Schreyer, and Zipperer (2017) and Monge-Naranjo, Sánchez, and Santaaulalia-Llopis (2019) and Manning and Barbier (2025) examines how the inclusion of natural capital inputs affects income-based growth accounting estimates of TFP. The results here show it is not possible to know whether these adjustments for non-renewable resources bring the resulting alternative estimates closer to the rate of true productivity growth over time unless they render the growth accounting equation correctly specified.

4 Simulating the Effects of Incorporating Natural Capital when Estimating TFP

Section 2 examined the theoretical effects of reducing the number of omitted factor inputs on the performance of the Solow residual as an estimator of TFP. This section evaluates this question using the available data on growth in factor inputs, natural capital stocks, and output by taking our running four-input economy to the data.

4.1 Setup

We consider a case where a national accountant seeks to measure TFP growth for this economy, but measures only growth in the use of produced capital, labor, and non-renewable capital, N_1 . We assume she suspects that services from renewable capital (N_2) are important inputs in production, but that she is unable to measure how their use has changed over time. Knowing this, she must choose whether to include growth in the use of non-renewable resources, g_{N_1} , when taking Equation (4) to the data to estimate TFP growth. Her criterion for including non-renewable resources in this thought experiment is whether it lowers expected in-sample root mean squared error when

estimating average TFP growth in her data, which we assume covers 25 years of growth in output and factor inputs.¹⁶

Data Generating Process for Output: Output growth in each country comprises growth in TFP, labor, produced capital, and two forms of natural capital inputs (non-renewables N_1 and renewables N_2). Joint observations of growth rates for all determinants of output growth are assumed to be independent and identically distributed draws of vectors $\mathbf{g}_{j,t}$ from a fixed distribution for each country with mean $\boldsymbol{\mu}_j$ and variance Σ_j . Output growth in each simulated country j is generated by

$$g_{Y,j,t} = \mathbf{g}_{j,t}' \boldsymbol{\gamma}_j \quad (22)$$

where $\boldsymbol{\gamma}_j \triangleq [1, \gamma_K, \gamma_L, \gamma_1, \gamma_2]'$ is the vector of output elasticities for country j .

Calibrating Country-Level Growth: We assume at the country level that growth in TFP (g_{TFP}) is uncorrelated with growth in other factors. We calibrate the mean ($\boldsymbol{\mu}_j$) and variance (Σ_j) governing the joint distribution of growth terms $\mathbf{g}_{j,t}$ in each country based on their analogs in the CWON and PWT. For natural capital, we follow the World Bank CWON methodology ([World Bank 2025](#)) and construct aggregate Törnqvist quantity indices for non-renewable and renewable resource use at the country level and set the parameters governing average growth in the use of natural capital equal to the sample means of growth in these indices between 1996 and 2019.¹⁷ Expected growth in produced capital services, labor, and TFP are set equal to their sample analogs in the Penn World Tables over the same 1996-2019 period.¹⁸ Table 3 below gives a full accounting of all data used in the simulation exercise along with the mapping between sample and population moments.

Output Elasticities: We calibrate the output elasticity of services from renewable natural capital, γ_2 , to equal 0.05 based on the analysis in Section 2.6 of the online appendix. In turn, to induce an overall output elasticity of 0.1 for natural capital (following Arrow et al. 2004), we set the output elasticity of services from non-renewable natural capital, γ_1 , to 0.05. We discuss sensitivity of our results to these choices of elasticities for natural capital in our results section below.

Procedure for Estimating TFP Growth: We assume the national accountant observes 25 years' worth of data on growth in factor inputs K, L , and N_1 and output Y in a given country. We assume she does not observe the distribution of national income earned across factors, and therefore cannot estimate output elasticities using factor shares. This ensures our thought experiment is

16. We round up to 25 from the size of 24 years in our estimating sample.

17. Section 2 of the online appendix outlines the full procedure for how we construct the Törnqvist indices for both forms of natural capital and calibrate $\boldsymbol{\mu}_j$ and Σ_j for each country.

18. Section 3.5.1 of the Appendix shows that re-estimating the sample moments used to calibrate country-level data generating processes based on two partitions of the original 1996-2019 data leads to larger estimates of RMSE reductions across the board and minimal qualitative changes in the set of countries for which ΔRMSE is negative.

consistent with estimating the Solow residual in empirical settings, where factor income generated by non-market natural capital services is unobservable to an econometrician (Manning, Taylor, and Wilen 2018).

To construct a measure of average TFP growth which does not rely on growth in non-renewable resources, the national accountant estimates:

$$g_Y = \hat{\gamma}_A + \hat{\gamma}_K g_K + \hat{\gamma}_L g_L + \epsilon \quad (23)$$

on time series of data for a given country using ordinary least squares (OLS). Her estimator for average TFP growth based only on growth in produced capital and labor is the intercept term $\hat{\gamma}_A$. We refer to this estimator as model 1. She then estimates the analog for the case including non-renewable resources:

$$g_Y = \tilde{\gamma}_A + \tilde{\gamma}_K g_K + \tilde{\gamma}_1 g_{N_1} + \tilde{\gamma}_L g_L + \nu \quad (24)$$

on time series for growth in produced capital, labor, and non-renewables. The OLS-based estimator for average TFP growth inclusive of non-renewable capital is $\tilde{\gamma}_A$. We refer to this estimator as model 2.

We then calculate the expected RMSE for $\hat{\gamma}_A$ and $\tilde{\gamma}_A$ in each country by evaluating models 1 and 2 at the sample analogs of $\boldsymbol{\mu}$ and Σ and comparing these estimates relative to the calibrated values, $\mathbb{E}[g_{TFP}]$:

$$\Delta \text{RMSE} | \boldsymbol{\mu}, \Sigma \triangleq \sqrt{\mathbb{E}[(\hat{\gamma}_A - \mathbb{E}[g_{TFP}])^2 | \boldsymbol{\mu}, \Sigma]} - \sqrt{\mathbb{E}[(\tilde{\gamma}_A - \mathbb{E}[g_{TFP}])^2 | \boldsymbol{\mu}, \Sigma]} \quad (25)$$

Applying this process to each country in the sample with sufficient data yields a cross section of 100 country-level observations of ΔRMSE .¹⁹

4.2 Simulation Results

For 94 out of the 100 countries for which we have sufficient data, including non-renewable resources (going from model 1 to model 2) lowers the RMSE of estimated TFP growth. Observations that fall in the green bins to the left of $x = 0$ in Figure 1 are countries where including non-renewable resources decreases the RMSE for a national accountant's residual-based estimate of TFP. The four observations in the blue bins to the right of $x = 0$ are countries for which including non-renewables increases the RMSE of estimated TFP growth (and decreases accuracy under this criterion). The median and mean declines in RMSE are sizable: reductions of 7.3 and 32 basis points (one-hundredth of a percentage point, abbreviated as bps) respectively. Reductions in error are dispersed (the standard deviation is 66 bps) and left-skewed. The countries with the largest reductions in RMSE are Kenya, Laos, Israel, Tanzania, and Niger. The magnitude of the median reduction is a little under 13 percent of average TFP growth across all countries over the 1996-2019

19. See Section 3.1 of the online appendix for the full procedure.

period. For the four countries where including non-renewables raises RMSE, the deterioration in fit is small (less than 1.2 basis points both on average and at the median). The greatest RMSE increase occurs for Chile, where RMSE rises by 4.8 bps.²⁰

Figure 2 shows scatter plots illustrating the cross-sectional variation in the relative performance of the estimators against individual country-level characteristics such as trends in resource use and baseline levels of calibrated productivity growth. Panels 2a and 2b plot the decline in mean squared error ($-\Delta\text{RMSE}_j$) induced by including non-renewables when estimating TFP growth against average growth in the use of non-renewable and renewable resources (g_{N_1} and g_{N_2}) at the country level. Of the 100 countries in our sample, 68 countries exhibit positive average growth in their use of non-renewables, and 47 exhibit positive average growth in the use of renewable inputs. Importantly, panels 2a and 2b show that average growth in non-renewables is about an order of magnitude greater than growth in renewables. This proportionally larger growth in non-renewable resources is responsible for substantially more output growth than that of renewables, even when the contribution of both resources to output in levels is similar. In turn, leaving out non-renewables on the margin leaves much more unexplained output in model 1 relative to model 2 where only renewables are omitted.

Panels 2c and 2d show point estimates of TFP growth from models 1 and 2 respectively against the true data generating process (set as the observed value in the PWT). Panels 2e and 2f show the changes in RMSE from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this change. The median (mean) decrease in RMSE is 10 (23) percent of baseline error when both natural capital stocks are omitted, and rises to more than 40 percent for over a quarter of the countries in the cross section of simulated data. Reduction in the bias of the estimators drives most of the changes in RMSE rather than a smaller expected variance. Bias drives 98 percent of RMSE on average when non-renewables are excluded (model 1), and 89 percent of bias on average when non-renewables are included (model 2).²¹

4.3 Δ RMSE in Context

The resulting decline in measurement error for most countries is of a magnitude that is policy-relevant. The median reduction in RMSE across the 100 countries in our sample is 7.3 basis points. This translates to an annual forecast error of roughly 0.07 percentage points when historical estimates are used to forecast average growth in the future. Applying this median value to the United States, when this forecast error is compounded over ten years (the standard CBO budget horizon) at current levels of output, forecast cumulative revenues would deviate from projections

20. Chile is an empirical case where the two components of bias in (23), stemming from omitting two variables, happen to offset each other almost perfectly, causing the bias when only renewables are omitted in (24) to be larger in magnitude.

21. See Section 3 of the online appendix for the derivation and decomposition of the changes in RMSE in terms of changes in bias and variance.

by approximately \$300 billion.²² This forecast error is roughly one quarter of the aggregate social cost of compliance with the U.S. Clean Air Act between 1970 and 1990, one of the broadest environmental policies enacted globally to date (U.S. Environmental Protection Agency 1997).

Decomposition across inputs: To decompose the contribution of individual components of output growth to changes in measurement accuracy, we regress ΔRMSE on the individual components of μ_j and Σ_j . This gives a reduced-form sense of the elements of input growth most important in determining the effects of measuring non-renewables.²³ Online appendix Table 16 shows that ΔRMSE is decreasing in the country-level growth rates of renewable and non-renewable resource use, and increasing in the correlations between growth in non-renewable resource use and those for labor and produced capital.²⁴ Thus if the world looks increasingly like Spain or Kyrgyzstan—nations where growth in the use of non-renewables is negatively correlated with produced capital and labor, then including non-renewable resources in national accounts is more likely to reduce the error in estimates of TFP growth. In contrast, if the world becomes more like Chile or Indonesia, where growth in the use of non-renewables is positively correlated with produced capital and labor, then adding non-renewables to national accounts will have a more limited impact on the error in existing estimates of TFP growth.²⁵

Other predictors of error reduction: The changes in RMSE resulting from including non-renewables do not appear correlated with other aggregate characteristics at the country level beyond moments governing growth in factor inputs. Figure 3 plots RMSE reductions against four additional statistics at the country level measured in 2019: (log) GDP, (log) GDP per capita, human capital levels, and statistical capacity.²⁶ While some relationships are statistically significant, the amount of variation in ΔRMSE explained by these factors is quite small (as indicated by the relative flatness of linear and non-parametric fits)., Panels 3a and 3b indicate that the magnitude of RMSE reductions tends to decrease in both measures of income. The relationship is robust and fairly substantial in magnitude; a three-fold increase in GDP decreases estimation error in TFP growth by 20 basis points (double our baseline estimate). A similar relationship holds between these reductions and the level of human capital in a given country (Panel 3c), which is unsurprising given the correlation between education and output. This relationship can be explained by higher-income countries having smaller and less-volatile TFP growth that lower the variance of both estimators

22. See Section 4 of the online appendix for a description of the back-of-the-envelope calculation reported here. The cumulative deviation for the United States would grow to slightly under \$1.2 trillion if we were to instead apply our U.S.-specific point estimate for the change in RMSE (26 bps) when including non-renewables on the margin.

23. The exact partial derivatives can also be calculated on a country-by-country basis using the closed-form solutions for ΔRMSE in Sections 3.2 and 3.3 of the online appendix.

24. Three of these four determinants of ΔRMSE are significant at traditional (5 percent) levels when ΔRMSE is regressed on the elements of μ_j and Σ_j that enter the RMSE calculations. The joint test is significant at the 0.1 percent level. See Section 3.4 of the online appendix for details and reduced-form analysis of how other sample moments determine ΔRMSE .

25. Figure 4 in Section 3.4 of the online appendix shows a version of these two panels for all 12 sample moments that determine ΔRMSE at the country level.

26. Statistical capacity is measured only for developing countries as defined by the World Bank.

and thus shrinks the error reduction in absolute terms.

Panel 3d shows the relationship between RMSE improvement and statistical capacity as measured by the World Bank for 67 developing countries in our sample. The correlation here is a fairly precise zero, which we interpret as suggestive evidence that better measurement of natural capital may benefit countries regardless of current national accounting practices. The last two panels take an alternative approach and examine how including non-renewables in our simulations relates to two measures of the *level* of resource intensity in production at the country-level. Panel 3e shows RMSE reductions against resource rents as share of GDP taken from the World Bank’s World Development Indicators database (World Bank 2024b). Panel 3f plots the improvements against an alternative metric for natural capital intensity, resource depletion as a share of gross national income. Perhaps surprisingly, there is a precise negative (albeit small in magnitude) statistical relationship between both measures of resource dependence and error reduction. For highly resource-dependent economies (e.g., Iraq and Kuwait), RMSE reductions are some of the smallest in our sample (1 and 3 basis points respectively). This negative relationship persists qualitatively after controlling for the other factors that have negative correlations shown here (i.e., human capital and GDP), but loses statistical significance and flips signs when conditioned on historical volatility in TFP (the driver of the variance portion of RMSE spoken to above).

Comparison with existing results: The magnitudes of ΔRMSE induced by including non-renewables here are similar to those produced by Brandt, Schreyer, and Zipperer (2017). The authors there find including non-renewable capital changes point estimates of annual TFP growth by an average of 2.3 basis points for a panel of 21 countries between 1986 and 2008. Manning and Barbier (2025) find adjusting the growth accounting equation to account for natural capital increases the amount of output growth unexplained by factor inputs by 87 basis points on average for countries with negative historical TFP growth and decreases it by 65 basis points on average for countries where TFP growth has been positive. Freeman, Inklaar, and Diewert (2021) use an earlier version of the COWON dataset to adjust TFP estimates to account for subsurface minerals and fossil fuels and find omitting non-renewable resources can bias TFP levels by up to 50 percent for resource-intensive economies.

Sensitivity analysis: We perform a Monte Carlo (MC) analysis to examine the sensitivity of our results to alternative pairs of output elasticities for the two natural capital stocks. Each individual MC draw produces an alternative ΔRMSE_j for a given country given a random pair of output elasticities for natural capital, γ_1, γ_2 , drawn from iid $\mathcal{U}[0, 0.1]$. We recalibrate the simulation ten million times by reassigning output elasticities based on these draws of alternative elasticity pairs at the country level. The median (mean) reduction in RMSE at the country level across our 10,000,000 alternative draws falls to 2 (28) basis points. The distribution remains quite disperse, displaying a standard deviation of 80 basis points. Unlike with our main analysis, a substantial share (31 percent) of country-level changes in RMSE induced by including non-renewables.

Figures 4a and 4b show the cross-sectional relationship between (average) reductions in RMSE at the country level and sampled values of output elasticities for non-renewable and renewable natural capital. All relationships shown are fit after winsorizing the Monte Carlo results of the Monte Carlo trial at the 90th percentile ($|\Delta\text{RMSE}_j| \geq 0.0085$) to avoid the leverage produced by observations in the left tail (i.e., from very large reductions in RMSE). Solid black lines show the fitted linear relationship between changes in (expected) ΔRMSE_j and the associated output elasticity γ_1 (γ_2) holding all else constant. Red dashed red lines show cubic splines between medians across 70 bins of γ_1 (γ_2). Dotted orange lines show local polynomial fits.

Figure 4a indicates that our findings that including non-renewables improves estimates TFP are least likely to hold if the output elasticity of non-renewable capital is on the lower end of the range we consider. All three fitted relationships (linear, binned median, and local polynomial) cross the x axis around $\gamma_1 = 0.015$, suggesting that (all else equal) including renewables when measuring TFP decreases accuracy. For resampled values less than the calibrated value in the main text (i.e., those where $\gamma_1 \leq 0.05$), 50.8% of Monte Carlo trials produce increases in RMSE (i.e., are cases where including non-renewables worsens measurement at the country level). Figure 4b repeats this exercise for the cross section of results by draws of γ_2 . For renewable resources, we find reduction in RMSE is least likely to hold when γ_2 is near the high end of values we consider. The median fit crosses the x -axis at around $\gamma = 0.075$. For resampled values larger than the calibrated value in the main text (i.e., those with $\gamma_2 \geq 0.05$), 45.9% of Monte Carlo trials produce negative improvements in RMSE (i.e., are cases where including non-renewables worsens measurement at the country level). Here, if the use of non-renewable natural capital begins to play a smaller role in production (the use of renewables becomes more important) holding all else equal, it is less likely that including it when estimating TFP will aid measurement.

5 Conclusion

Fifty-four years after Nordhaus and Tobin (1973), research continues to highlight new channels by which environmental change affects society and welfare. An increasing share of this research has focused on responding to high-level calls for improved welfare measures, e.g., the U.N. Secretary General’s push for measures “Beyond GDP” (United Nations Statistics Division 2026). Economists have long recognized the importance of such measures, and there is a rich literature on complements to GDP that would better account for the determinants of welfare beyond market activity (Fleurbaey 2009; Jorgenson 2018).²⁷ This paper gives evidence that measurement of natural capital benefits governments irrespective of goals beyond GDP. When natural capital affects production, omitting it from national accounts biases estimates of economic growth. This supports the original hypothesis by Nordhaus and Tobin in 1973 that leaving things out (“treating as free things that are not really free”) of national accounts “... gives the wrong signals for the directions of economic growth” in addition to affecting measurement of national welfare (Nordhaus and Tobin 1973). While

27. Conte, Addicott, and Millerhaller (2025) provide micro-founded support for this position regarding the environment, demonstrating how reductions in natural capital stocks can generate real income effects for consumers.

measuring the environment will remain important for assessing the sustainability of development, regulatory compliance, and public health, we show here that systematically connecting changes in the environment to economic statistics will benefit existing measurement that is already used to guide policy.

6 Proofs

6.1 Proposition 1

Proof. We calculate the bias for the estimator in Model 1 conditional on estimated output elasticities for capital and labor $\{\hat{\gamma}_K, \hat{\gamma}_L\}$. Note that with CRS production and competitive markets, estimated elasticities for the two-factor estimator must satisfy $\hat{\gamma}_K + \hat{\gamma}_L = 1$. Proposition 1 can then be proved in parts.

Part 1: Let $\bar{g} \triangleq \max\{\mathbb{E}[g_K], \mathbb{E}[g_L]\}$ and $\underline{g}_N \triangleq \min_j\{\mathbb{E}[g_{N_j}]\}$. Suppose that $\underline{g}_N > \bar{g}$. Then

$$\begin{aligned} \mathbb{E}[\varepsilon|\hat{\gamma}_K, \hat{\gamma}_L] &= [\gamma_K - \hat{\gamma}_K]\mathbb{E}[g_K] + [\gamma_L - (1 - \hat{\gamma}_K)]\mathbb{E}[g_L] + \sum_{j=1}^J \gamma_j \mathbb{E}[g_{N_j}] \\ &\geq \sum_{j=1}^J \gamma_j \mathbb{E}[g_{N_j}] + [\gamma_K]\mathbb{E}[g_K] + [\gamma_L]\mathbb{E}[g_L] - \bar{g} \\ &\geq \sum_{j=1}^J \gamma_j \mathbb{E}[g_{N_j}] + [\gamma_K + \gamma_L]\bar{g} - \bar{g} \\ &= \sum_{j=1}^J \gamma_j \mathbb{E}[g_{N_j}] - \sum_{j=1}^J \gamma_j \bar{g} \end{aligned}$$

where the first and second inequalities follow from the definition of $\bar{g}$ and the last equality follows from $1 - \gamma_K - \gamma_L = \sum_{j=1}^J \gamma_j$. We then have

$$\mathbb{E}[\varepsilon|\hat{\gamma}_K, \hat{\gamma}_L] \geq [\underline{g}_N - \bar{g}] \sum_{j=1}^J \gamma_j \geq 0$$

as $\underline{g}_N - \bar{g} > 0$ by assumption. Finally, note that adding a previously-omitted factor input term lowers bias, as we reduce the number of positive factor shares γ_j in the summation in the last line by increasing the starting index to $j = 2$ (without loss of generality).

Part 2: Let $\underline{g} \triangleq \max\{\mathbb{E}[g_K], \mathbb{E}[g_L]\}$ and $\bar{g}_N \triangleq \min_j\{\mathbb{E}[g_{N_j}]\}$ and suppose that $\underline{g} > \bar{g}_N$. Then analogous to the presentation above we have

$$\begin{aligned} \mathbb{E}[\varepsilon|\hat{\gamma}_K, \hat{\gamma}_L] &= [\gamma_K - \hat{\gamma}_K]\mathbb{E}[g_K] + [\gamma_L - (1 - \hat{\gamma}_K)]\mathbb{E}[g_L] + \sum_{j=1}^J \gamma_j \mathbb{E}[g_{N_j}] \\ &\leq \sum_{j=1}^J \gamma_j \mathbb{E}[g_{N_j}] - \sum_{j=1}^J \gamma_j \underline{g} \\ &\leq [\bar{g}_N - \underline{g}] \sum_{j=1}^J \gamma_j \leq 0 \end{aligned}$$

where in this case adding a correctly-measured omitted input stock (removing a term from the summation) decreases bias in nominal and absolute terms.

6.2 Corollary 1

Part 1 In the general case, the bias of Model 1 is given by

$$\mathbb{E}[\varepsilon|\hat{\gamma}_K, \hat{\gamma}_L] = [\gamma_K - \hat{\gamma}_K]\mathbb{E}[g_K] + [\gamma_L - (1 - \hat{\gamma}_K)]\mathbb{E}[g_L] + \sum_{j=1}^J \gamma_j \mathbb{E}[g_{N_j}]$$

where we've substituted $\hat{\gamma}_L = 1 - \hat{\gamma}_K$. The sign of the right-hand side depends on the expected values of omitted growth terms $\mathbb{E}[g_{N_j}]$ which are unknown to the econometrician and may be positive or negative. Even if all growth terms are assumed to be a single sign (weakly positive or negative) in expectation, the sign of the bias will still depend on the relative magnitudes of output elasticities γ .

Part 2 Suppose now that one of the output elasticities for the included growth terms is measured without error such that $\hat{\gamma}_K = \gamma_K$. Under the assumption of CRS with competitive markets, the labor share is then $(1 - \hat{\gamma}_K)$. Then the bias from the first model is:

$$\mathbb{E}[\varepsilon|\hat{\gamma}_K, \hat{\gamma}_L] = \sum_{j=1}^J \gamma_j \mathbb{E}[g_L - g_{N_j}] \leq 0$$

Alternatively, if $\hat{\gamma}_L = \gamma_L$ (the labor share is measured without error), then we have $\hat{\gamma}_K = \gamma_K + \sum_j \gamma_j$ which yields

$$\mathbb{E}[\varepsilon|\hat{\gamma}_K, \hat{\gamma}_L] = \sum_{j=1}^J \gamma_j \mathbb{E}[g_K - g_{N_j}] \leq 0$$

Part 3 Suppose instead that an econometrician is able to measure output elasticities for both included factors without error. While this causes the bias to comprise only the omitted factor terms, its sign remains ambiguous:

$$\mathbb{E}[\varepsilon|\hat{\gamma}_K, \hat{\gamma}_L] = \sum_{j=1}^J \gamma_j \mathbb{E}[g_{N_j}] \leq 0$$

□

6.3 Proposition 2

Proof. We prove the result in Proposition 2 here the result for a special case where each input growth term is iid and output elasticities and factor input growth are known to the econometrician without error. Introducing correlations in growth terms or measurement error adds more potential sources of ambiguity, but is not necessary to show that adding additional natural capital stocks

has ambiguous effects on bias. In the general case, suppose Model 1 includes a subset of inputs $\mathbf{I} \subset \mathbf{X}$ and that all associated output elasticities $\hat{\gamma}$. Suppose Model 2 includes the subset $\mathbf{I}'$ with associated output elasticities $\tilde{\gamma}$ where $\mathbf{I} \subset \mathbf{I}' \subset \mathbf{X}$. The biases are in general

$$\begin{aligned}\mathbb{E}[\varepsilon|\hat{\gamma}] &= \sum_{i \in \mathbf{I}} [\gamma_i - \hat{\gamma}_i] \mathbb{E}[g_{N_i}] + \sum_{j \in \mathbf{X} \setminus \mathbf{I}} \gamma_j \mathbb{E}[g_{N_j}] \\ \mathbb{E}[\delta|\tilde{\gamma}] &= \sum_{i \in \mathbf{I}'} [\gamma_i - \tilde{\gamma}_i] \mathbb{E}[g_{N_i}] + \sum_{j \in \mathbf{X} \setminus \mathbf{I}'} \gamma_j \mathbb{E}[g_{N_j}]\end{aligned}$$

and for the specific case when output elasticities are estimated without error in both models, the above simplifies to

$$\begin{aligned}\mathbb{E}[\varepsilon|\hat{\gamma}] &= \sum_{j \in \mathbf{X} \setminus \mathbf{I}} \gamma_j \underbrace{\mathbb{E}[g_{N_j}]}_{\leq 0} \\ \mathbb{E}[\delta|\tilde{\gamma}] &= \sum_{j \in \mathbf{X} \setminus \mathbf{I}'} \gamma_j \underbrace{\mathbb{E}[g_{N_j}]}_{\leq 0}\end{aligned}$$

and the magnitude and sign of both biases remains indeterminate.

The mean squared errors of our estimators when all factor shares and growth terms are measured without error simplify to the squared value of the biases. Note that

$$\begin{aligned}\text{MSE}(1)|\hat{\gamma} &= \mathbb{E} \left[\left(\sum_{j \in \mathbf{X} \setminus \mathbf{I}} \gamma_j \mathbb{E}[g_{N_j}] \right)^2 \right] \\ \text{MSE}(2)|\tilde{\gamma} &= \mathbb{E} \left[\left(\sum_{j \in \mathbf{X} \setminus \mathbf{I}'} \gamma_j \mathbb{E}[g_{N_j}] \right)^2 \right]\end{aligned}$$

Evaluating both means squared errors gives

$$\begin{aligned}\text{MSE}(1)|\hat{\gamma} &= \mathbb{E} \left[\left(\sum_{j \in \mathbf{X} \setminus \mathbf{I}} \gamma_j \mathbb{E}[g_{N_j}] \right)^2 \right] \\ &= \text{Var} \left(\sum_{j \in \mathbf{X} \setminus \mathbf{I}} \gamma_j \mathbb{E}[g_{N_j}] \right) + \mathbb{E}[\varepsilon|\hat{\gamma}]^2 \\ &= \mathbb{E}[\varepsilon|\hat{\gamma}]^2\end{aligned}$$

where the last line follows because the γ_j and $\mathbb{E}[g_{N_j}]$ terms are not random variables. The analog for Model 2 is

$$\text{MSE}(2)|\tilde{\gamma} = \mathbb{E}[\delta|\tilde{\gamma}]^2$$

The difference in mean squared error is then

$$\text{MSE}(1) - \text{MSE}(2) = \underbrace{\mathbb{E}[\varepsilon|\tilde{\gamma}]^2 - \mathbb{E}[\delta|\tilde{\gamma}]^2}_{\leq 0} \leq 0$$

where the inequality is ambiguous unless $\mathbb{E}[\delta] = 0$ in the case where $\mathbf{I} = \mathbf{X}$ and Model 2 is correctly specified. This ambiguity generalized to the case where the measurement error in the estimated output elasticities is treated as a random variable as extra terms of ambiguous sign enter the MSE formulas in the form of covariances between estimated output elasticities under each model. $\square$

7 Tables and Figures

Table 1: Renewable natural capital stocks quantified in the Changing Wealth of Nations

Resource	Natural Unit	Instrumented Service Flow	Potential Confounders	% Obs. Missing	# Countries
Timber land	Hectares	Timber harvest, carbon capture	Decline in seeding services	1.25%	114
Agricultural land	Square kilometers	Grazing area, soil services	Decline in ecosystem services from fallow land	2.56%	116
Non-timber forest cover	Square kilometers	Recreation, ecosystem services	Decline in grazing area and forestry	3.95%	113
Mangrove area	Hectares	Flood protection, water filtration, embedded habitats	Marine recreation	63.25%	43
Fishery biomass	Metric tons	Fished biomass	Nutrient cycling, pump dynamics	23.08%	90
Hydroelectric generation	Gigawatt-hours	Electricity, flood prevention, water storage	Ecosystem services from rivers and floodplains	27.07%	86

Note: All data here are taken from the World Bank *Changing Wealth of Nations* database ([World Bank 2025](#)). The first two columns report each resource and the unit in which physical quantities are measured. The column labeled “Instrumented Service Flow” gives an example of services from natural capital that we believe affect TFP that are likely correlated with the change in these natural capital stocks. The “Potential Confounders” column reports how this stock may also be correlated with (1) other omitted inputs to GDP beyond natural capital of (2) measurement error for the factors that are included. being omitted from standard national accounts. The column “% obs. missing” reports the share of observations between 1996 and 2019 with missing quantities in the initial dataset. The “# Countries” column shows the number of countries where at least one observation is non-zero.

Table 2: TFP Growth vs. Growth in Renewable Natural Capital Stocks

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Timber Land	0.164 (0.049)	0.166 (0.046)	0.071 (0.048)	0.075 (0.046)	-0.003 (0.053)	0.026 (0.104)
Agricultural Land	0.027 (0.019)	0.043 (0.019)	0.030 (0.019)	0.036 (0.019)	0.041 (0.019)	0.043 (0.017)
Forest Cover	-0.067 (0.061)	-0.111 (0.064)	-0.005 (0.060)	0.012 (0.061)	0.083 (0.055)	-0.027 (0.076)
Mangroves	0.122 (0.128)	0.067 (0.131)	0.357 (0.163)	0.327 (0.167)	0.318 (0.202)	-0.108 (0.427)
Fishery Biomass	0.112 (0.056)	0.026 (0.054)	0.094 (0.064)	0.072 (0.061)	0.047 (0.083)	-0.104 (0.108)
Hydropower	0.011 (0.005)	0.011 (0.005)	0.009 (0.005)	0.009 (0.005)	0.006 (0.006)	0.006 (0.003)
Controls for . . .						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
Wald	0.001	0.001	0.016	0.017	0.090	0.045
Bootstrap Wald	0.001	0.002	0.017	0.016	–	0.061
# Observations	2,808	2,808	2,808	2,808	2,808	2,457
# Countries	117	117	117	117	117	117

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables (Feenstra, Inklaar, and Timmer 2015). All independent variables enter as log changes. Robust standard errors are shown in parentheses. Rows under the “Controls for . . .” heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator and reports robust (as opposed to clustered) standard errors. Row “Wald” displays p -values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row “Bootstrap Wald” shows p -values from wild bootstrap tests of the joint null hypothesis (Wu 1986). No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Table 3: Simulation Moments and Data Sources

Description	Population Moment	Sample Moment	Source
Non-Renewable Services	$\mathbb{E}[g_{N_1}]$	$\bar{g}^R$	CWON
Renewable Services	$\mathbb{E}[g_{N_2}]$	$\bar{g}^{NR}$	CWON
Produced Capital Services	$\mathbb{E}[g_K]$	$\bar{g}_K$	PWT
Labor Inputs	$\mathbb{E}[g_L]$	$\bar{g}_L$	PWT
TFP	$\mathbb{E}[g_{TFP}]$	$\bar{g}_{TFP}$	PWT
Covariances	Σ	$\hat{\Sigma}$	CWON & PWT

Note: Subscripts indexing each object by country are dropped in the first two columns; all population moments are assigned the sample analogs on a country-by-country basis. Bars denote sample averages while hats denote sample analogs for the covariance matrix Σ . CWON refers to the World Bank *Changing Wealth of Nations* database ([World Bank 2025](#)). PWT refers to the Penn World Tables version 10.01 (Feenstra, Inklaar, and Timmer 2015).

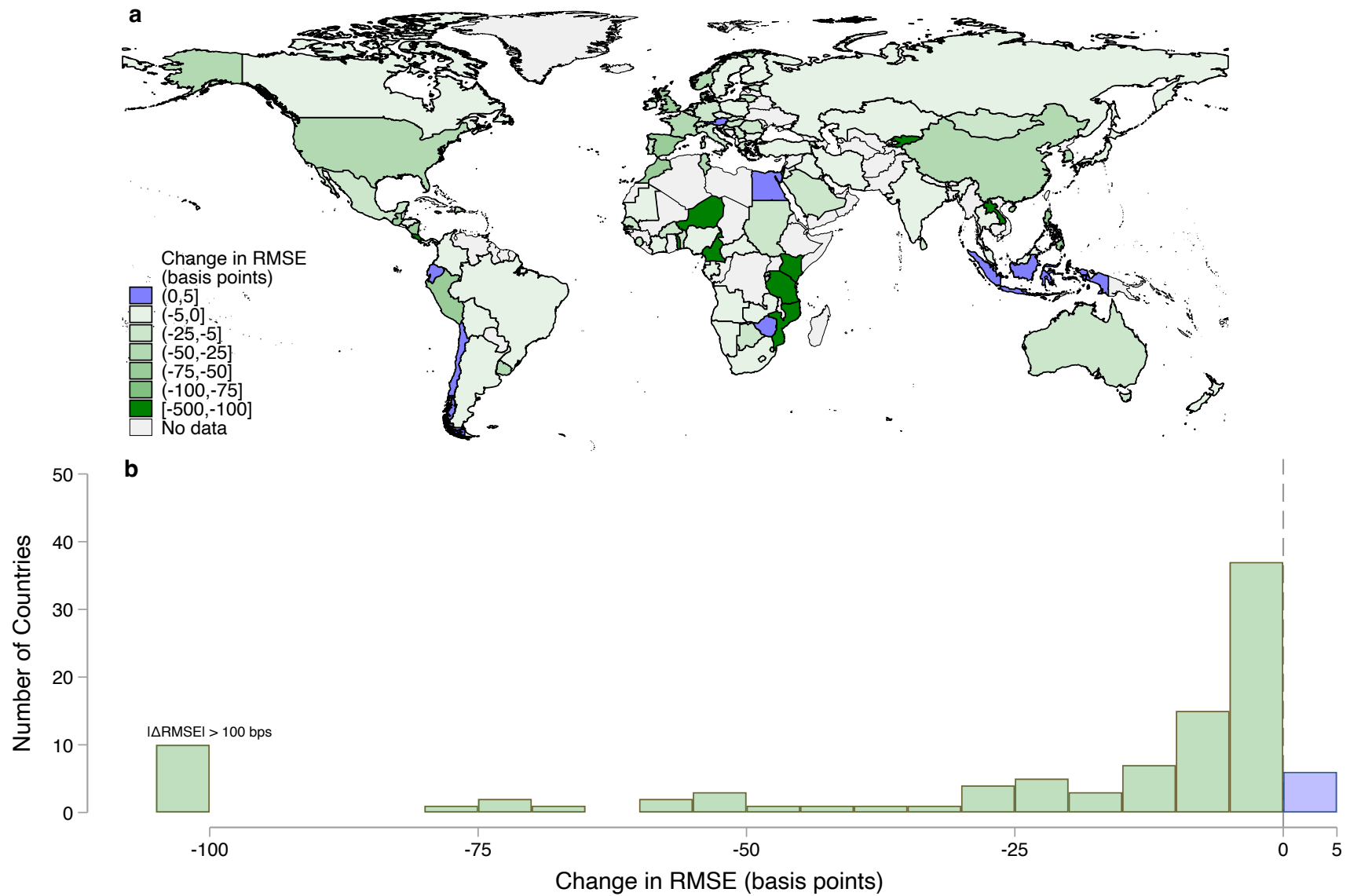


Figure 1: Panel **a**. Map of ΔRMSE for the 100 countries in our sample. Panel **b**. Histogram of the reductions in root mean squared error— ΔRMSE —when adding g_{N_1} to the Solow residual-based TFP estimator for each of 100 countries in the sample. The 10 countries with RMSE reductions larger than 100 basis points in magnitude are topcoded and added to the leftmost bin in the histogram. See Table 15 in the online appendix for a full list of country-level sample moments and changes in RMSE.

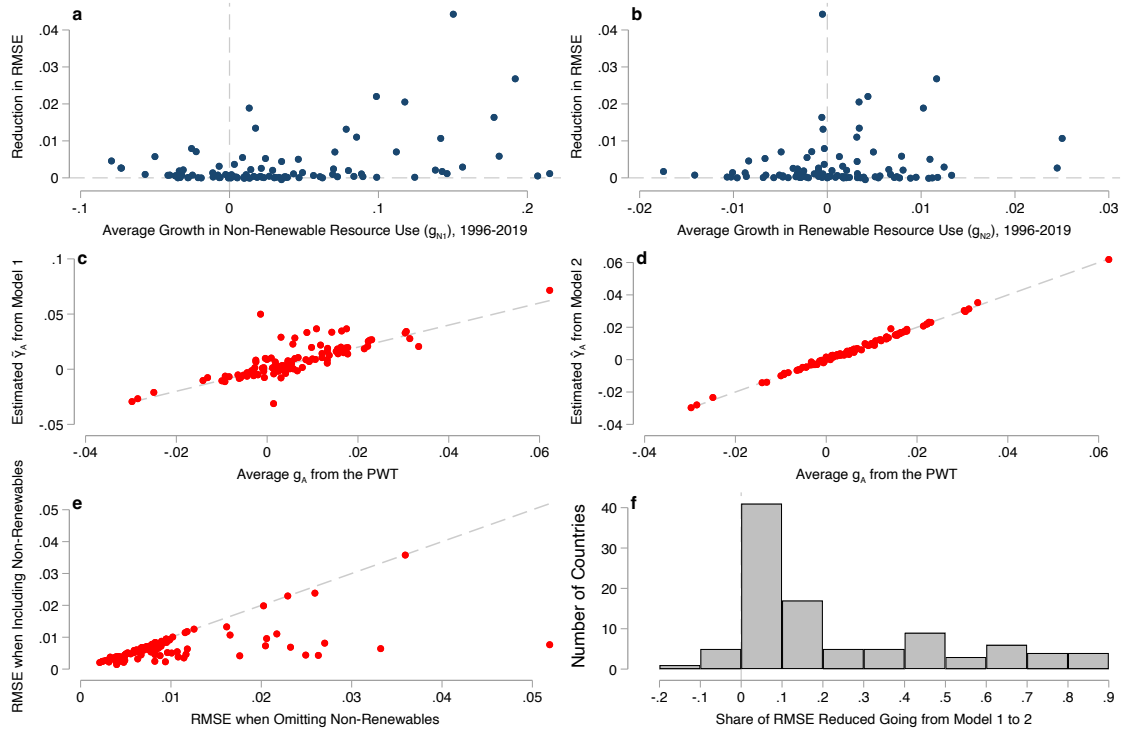


Figure 2: Panels 2a and 2b plot ΔRMSE when estimating TFP growth against average growth in non-renewable and renewable resource use at the country level between 1996 and 2019 for our panel of 100 countries. Panels 2c and 2d show point estimates of TFP growth based on historical moments from the two estimators against the true data generating process. Panels 2e and 2f show the changes in RMSE from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this change. Dashed lines in panels 2a, 2b, and 2f are plotted at $x = 0$ and $y = 0$. Dashed lines in panels 2c-2e show the 45 degree line.

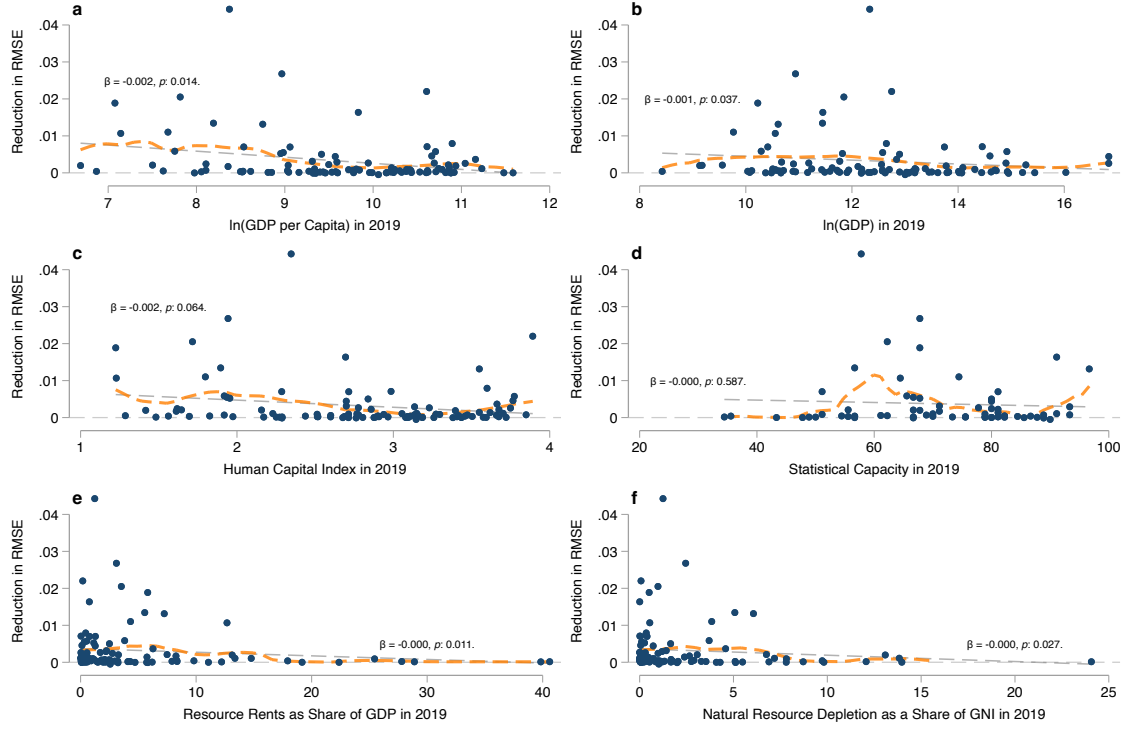
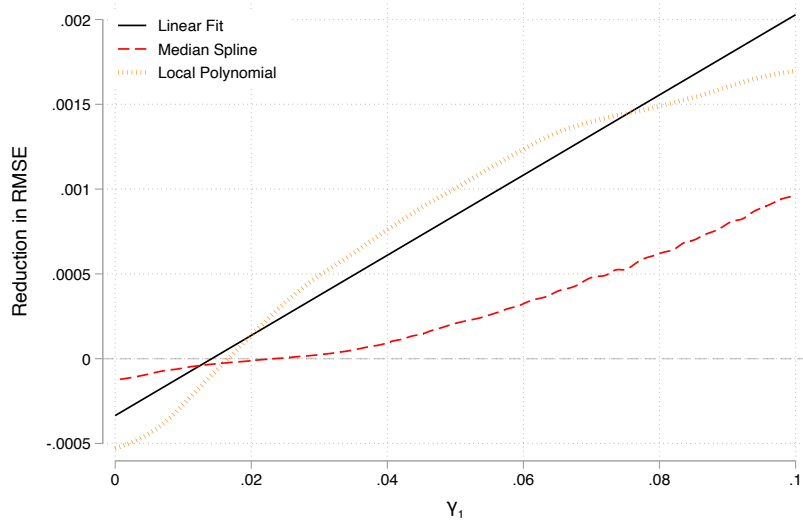
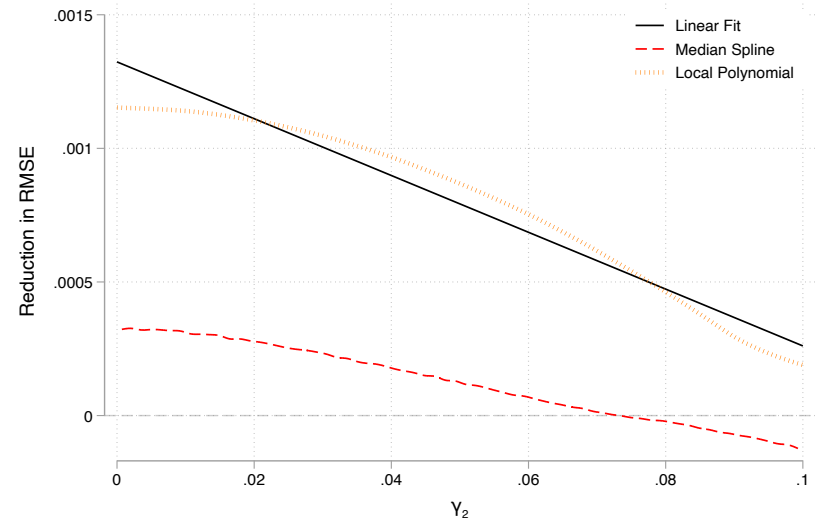


Figure 3: Panels 3a and 3b plot reductions in the RMSE of estimated TFP growth against the (logarithm) of GDP per capita and aggregate GDP at the country level. Panels 3c and 3d plots these improvements against measures of human capital and statistical capacity take from the Penn World Tables (Feenstra, Inklaar, and Timmer 2015). Panels 3e and 3f show RMSE reductions against the share of GDP that comes from resource rents and the share of national income attributable to natural resource depletion, respectively. Dashed lines show fits from a simple linear regressions and the overlain text shows the $\hat{\beta}$ coefficient from the linear regression along with the associated p -values. Dashed orange lines shows fitted values from a local linear regression (Fan 2018).



(a) Sensitivity of ΔRMSE to Calibrated γ_1



(b) Sensitivity of ΔRMSE to Calibrated γ_2

Figure 4: Linear, median spline, and local polynomial fit lines for ΔRMSE_j at the country level against draws of γ_1 and γ_2 across $N = 10,000,000$ Monte Carlo draws. Fits are performed after winsorizing draws at the 90th percentile ($|\Delta\text{RMSE}_j| \geq 0.0085$) to reduce the leverage of the right tail of the ΔRMSE_j distribution. The median spline fit is generated by a cubic spline between median values of ΔRMSE_j across observations within each of 70 intervals for the relevant parameter.

References

- Acemoglu, Daron, Philippe Aghion, Lint Barrage, and David Hémous. 2023. *Climate change, directed innovation, and energy transition: The long-run consequences of the shale gas revolution*. Technical report. National Bureau of Economic Research. (Cited on page 7).
- Aghion, Philippe, and Peter Howitt. 1992. “A Model of Growth through Creative Destruction.” *Econometrica* 60 (2): 323–351. (Cited on page 3).
- Anderson, Soren T, Ryan Kellogg, and Stephen W Salant. 2018. “Hotelling under pressure.” *Journal of Political Economy* 126 (3): 984–1026. (Cited on page 5).
- Arrow, Kenneth, Partha Dasgupta, Lawrence Goulder, Gretchen Daily, Paul Ehrlich, Geoffrey Heal, Simon Levin, Karl-Göran Mäler, Stephen Schneider, David Starrett, et al. 2004. “Are we consuming too much?” *Journal of Economic Perspectives* 18 (3): 147–172. (Cited on page 17).
- Barro, Robert J. 1999. “Notes on growth accounting.” *Journal of economic growth* 4 (2): 119–137. (Cited on page 3).
- Basu, Susanto, Luigi Pascali, Fabio Schiantarelli, and Luis Servén. 2022. “Productivity and the Welfare of Nations.” *Journal of the European Economic Association* 20 (4): 1647–1682. (Cited on page 1).
- Belloni, Alexandre, Victor Chernozhukov, and Christian Hansen. 2014. “Inference on treatment effects after selection among high-dimensional controls.” *Review of Economic Studies* 81 (2): 608–650. (Cited on page 12).
- Brandt, Nicola, Paul Schreyer, and Vera Zipperer. 2017. “Productivity measurement with natural capital.” *Review of Income and Wealth* 63:S7–S21. (Cited on pages 1, 2, 15, 16, 21).
- Bunting, Pete, Ake Rosenqvist, Lara Hilarides, Richard M. Lucas, Tom Thomas, Takeo Tadono, Thomas A. Worthington, Mark Spalding, Nicholas J. Murray, and L-M. Rebelo. 2022. “Global Mangrove Extent Change 1996–2020: Global Mangrove Watch Version 3.0.” *Remote Sensing* 14 (15): 3657. <https://doi.org/10.3390/rs14153657>. (Cited on page 10).
- Caselli, Francesco, and James Feyrer. 2007. “The marginal product of capital.” *The quarterly journal of economics* 122 (2): 535–568. (Cited on pages 2, 7, 14, 15).
- CBO. 2025. *The Budget and Economic Outlook: 2025 to 2035*. Report No. 60870. Accessed 2026-01-19. Congressional Budget Office. <https://www.cbo.gov/publication/60870>. (Cited on page 1).
- . 2026. *The Accuracy of CBO’s Budget Projections for Fiscal Year 2025*. 61916. Congressional Budget Office, January. <https://www.cbo.gov/system/files/2026-01/61916-budget-accuracy.pdf>. (Cited on page 2).

- Comin, Diego. 2010. “Total factor productivity.” In *Economic growth*, 260–263. Springer. (Cited on page 4).
- Conte, Marc N., Ethan T. Addicott, and Maxanne M. Millerhaller. 2025. “Scarcity, willingness to pay for species, and imperfect substitutability with market goods.” *Journal of Environmental Economics and Management*, 103192. (Cited on page 22).
- European Commission and Eurostat. 2025. “Multi-Factor Productivity (MFP) Using National Accounts Data: Methodological Note.” <https://ec.europa.eu/eurostat/documents/7894008/13933430/Methodological-note-crude-MFP.pdf>. (Cited on page 4).
- Fan, Jianqing. 2018. *Local polynomial modelling and its applications: monographs on statistics and applied probability* 66. Routledge. (Cited on page 34).
- Feenstra, Robert C., Robert Inklaar, and Marcel P. Timmer. 2015. “The Next Generation of the Penn World Table.” *American Economic Review* 105 (10): 3150–3182. (Cited on pages 4, 7, 8, 10, 12, 30, 31, 34).
- Fleurbaey, Marc. 2009. “Beyond GDP: The quest for a measure of social welfare.” *Journal of Economic literature* 47 (4): 1029–1075. (Cited on page 22).
- Freeman, Daan, Robert Inklaar, and W. Erwin Diewert. 2021. “Natural resources and missing inputs in international productivity comparisons.” *Review of Income and Wealth* 67 (1): 1–17. (Cited on pages 3, 21).
- Hannesson, Rögnvaldur. 2007. “Growth accounting in a fishery.” *Journal of environmental economics and management* 53 (3): 364–376. (Cited on pages 1, 5).
- Holmlund, Cecilia M, and Monica Hammer. 1999. “Ecosystem services generated by fish populations.” *Ecological economics* 29 (2): 253–268. (Cited on page 6).
- Hsieh, Chang-Tai, and Peter J. Klenow. 2010. “Development Accounting.” *American Economic Journal: Macroeconomics* 2 (1): 207–223. (Cited on page 2).
- Jorgenson, Dale W. 2018. “Production and welfare: progress in economic measurement.” *Journal of Economic Literature* 56 (3): 867–919. (Cited on page 22).
- Kuznets, Simon. 1934. *National Income, 1929–1932*. 73rd Congress, 2nd Session, Senate Document No. 124. Washington, DC: U.S. Government Printing Office. (Cited on page 1).
- Lemoine, Derek, Catherine Hausman, and Jeffrey G Shrader. 2026. “Navigating the ‘Problem from Hell’: A Guide to Climate Damages.” *Journal of Economic Literature*, (cited on page 9).
- Mankiw, N. Gregory, David Romer, and David N. Weil. 1992. “A contribution to the empirics of economic growth.” *The quarterly journal of economics* 107 (2): 407–437. (Cited on pages 2, 14).

- Manning, Dale T, and Edward B Barbier. 2025. “Natural capital and aggregate income growth.” *Environment and Development Economics* 30 (2): 155–174. (Cited on pages 2, 15, 16, 21).
- Manning, Dale T., J. Edward Taylor, and James E. Wilen. 2018. “General equilibrium tragedy of the commons.” *Environmental and Resource Economics* 69 (1): 75–101. (Cited on page 18).
- Meyer, Peter B., and Michael J. Harper. 2005. “Preliminary Estimates of Multifactor Productivity Growth.” *Monthly Labor Review* (June). <https://www.bls.gov/opub/mlr/2005/06/art3full.pdf>. (Cited on page 4).
- Monge-Naranjo, Alexander, Juan M Sánchez, and Raül Santaaulalia-Llopis. 2019. “Natural resources and global misallocation.” *American Economic Journal: Macroeconomics* 11 (2): 79–126. (Cited on pages 2, 14–16).
- Moscona, Jacob, and Karthik A Sastry. 2023. “Does directed innovation mitigate climate damage? Evidence from US agriculture.” *The Quarterly Journal of Economics* 138 (2): 637–701. (Cited on page 7).
- Nordhaus, William D, and Edward C Kokkelenberg. 1999. *Nature’s Numbers: Expanding the National Economic Accounts to Include the Environment*. (Cited on page 4).
- Nordhaus, William D., and James Tobin. 1973. “Is Growth Obsolete?” In *The Measurement of Economic and Social Performance*, edited by Milton Moss, 509–564. National Bureau of Economic Research. (Cited on pages 1, 2, 22).
- Organisation for Economic Co-operation and Development. 2001. *Measuring Productivity: OECD Manual*. Paris: OECD Publishing. <https://unstats.un.org/unsd/nationalaccount/docs/OECD-Productivity-e.pdf>. (Cited on page 4).
- Pauly, Daniel, Dirk Zeller, and Maria L. D. Palomares. 2020. *Sea Around Us Concepts, Design and Data*. <https://www.seaaroundus.org>. Accessed April 5, 2026. (Cited on page 10).
- Romer, Paul M. 1990. “Endogenous Technological Change.” *Journal of Political Economy* 98 (5, pt. 2): S71–S102. (Cited on page 3).
- Roosevelt, Theodore. 1908. *Conservation as a National Duty*. Speech delivered at the Governors’ Conference on the Conservation of Natural Resources, Washington, DC, May 13–15; reprinted in *Messages and Papers of the Presidents*, vol. 7 (1909), Bureau of National Literature. (Cited on page 1).
- Solow, Robert M. 1957. “Technical change and the aggregate production function.” *The review of Economics and Statistics* 39 (3): 312–320. (Cited on pages 1, 4).

- U.S. Environmental Protection Agency. 1997. *The Benefits and Costs of the Clean Air Act, 1970 to 1990: Report to Congress*. Technical report EE-0295 (Retrospective Report). Accessed 2026-02-02. U.S. Environmental Protection Agency, Office of Air and Radiation. <https://www.epa.gov/sites/default/files/2015-06/documents/contsetc.pdf>. (Cited on pages 2, 20).
- United Nations. 2009. *System of National Accounts 2008*. Technical report. European Commission, International Monetary Fund, Organisation for the Economic Co-operation and Development, United Nations. (Cited on page 1).
- . 2025. *System of National Accounts 2025*. Technical report. European Commission, International Monetary Fund, Organisation for the Economic Co-operation and Development, United Nations. (Cited on pages 1, 4).
- United Nations Statistics Division. 2026. *Beyond GDP: Rethinking Progress in a Changing World — Concept Note*. United Nations Statistics Division, March. https://unstats.un.org/UNSDWebsite/statcom/session_57/side-events/Monday_Seminar_2026_Final_Beyond_GDP_Concept_Note.pdf. (Cited on page 22).
- World Bank. 2024a. *The Changing Wealth of Nations 2024: Methodology*. Technical Report. Accessed April 5, 2026. World Bank, October. https://datacatalogfiles.worldbank.org/ddh-published/0042066/10/DR0094582/CWON%202024%20Methodology_10122024.pdf. (Cited on page 10).
- . 2024b. *World Development Indicators*. <https://databank.worldbank.org/source/world-development-indicators>. Accessed 2026-03-13. World Bank, Washington, DC. (Cited on page 21).
- . 2025. *The Changing Wealth of Nations (CWON) Dataset*. https://data360.worldbank.org/en/dataset/WB_CWON. (Cited on pages 10, 17, 29, 31).
- Wu, Chien-Fu Jeff. 1986. “Jackknife, bootstrap and other resampling methods in regression analysis.” *The Annals of Statistics* 14 (4): 1261–1295. (Cited on pages 11, 30).